

# THE STRANGE COUSIN OF COMPLEX NUMBERS

$\mathbb{R}(\varepsilon)$  vs.  $\mathbb{C}$

Definition: The dual numbers  $\mathbb{R}(\varepsilon)$  are all numbers of

Clifford  
1873

the form

$$a + b\varepsilon$$

where  $a, b \in \mathbb{R}$  and  $\varepsilon^2 = 0$ .

\* Compare to complex numbers  $\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}, i^2 = -1\}$

Some Example Calculations

product:  $(2 + 3\varepsilon)(5 - \varepsilon) = 10 + 13\varepsilon - \cancel{3\varepsilon} = 10 + 13\varepsilon$

$$(a + b\varepsilon)(c + d\varepsilon) = ac + (ad + bc)\varepsilon$$

What are the zero divisors of the Ring  $\mathbb{R}(\varepsilon)$ ?

i.e. all  $a + b\varepsilon, c + d\varepsilon \neq 0$  such that  $0 = (a + b\varepsilon)(c + d\varepsilon)$   
 $= ac + (ad + bc)\varepsilon$

$$\Rightarrow ac = 0 \text{ and } ad + bc = 0$$

$$\text{case 1: } a = 0 \Rightarrow bc = 0 \Rightarrow c = 0 \Rightarrow (b\varepsilon)(d\varepsilon) = 0$$

$$\text{case 2: } c = 0 \Rightarrow ad = 0 \Rightarrow a = 0 \Rightarrow$$

$\rightarrow$  zero divisors are  $b\varepsilon, b \neq 0$

quotients:  $\frac{3 + 2\varepsilon}{2 + 5\varepsilon} \cdot \frac{(2 - 5\varepsilon)}{(2 - 5\varepsilon)} = \frac{6 - 11\varepsilon}{4} = \frac{3}{2} - \frac{11}{4}\varepsilon$

$$\frac{a + b\varepsilon}{c + d\varepsilon} = \frac{ac + (bd - ad)\varepsilon}{c^2}$$

## Relation to the Derivative

polynomial  $P(x) = x^2$

$$P(x+\varepsilon) = (x+\varepsilon)^2 = x^2 + 2x\varepsilon = P(x) + P'(x) \cdot \varepsilon$$

$$\Rightarrow P(x+\varepsilon) - P(x) = P'(x) \cdot \varepsilon$$

Now we can't divide with  $\varepsilon$ , 'cause  $1 \cdot \varepsilon$  is a zero divisor in the ring  $\mathbb{R}(\varepsilon)$ .

In general:  $f(x) = \sum_{n=0}^{\infty} a_n x^n$

$$f(x+\varepsilon) = \sum_{n=0}^{\infty} a_n (x+\varepsilon)^n = \sum_{n=0}^{\infty} \sum_{k=0}^n a_n \binom{n}{k} x^{n-k} \varepsilon^k$$

$\swarrow$   $= 0, 1$   
(not is 0)

$$= \underbrace{\sum_{n=0}^{\infty} a_n x^n}_{k=0} + \varepsilon \underbrace{\sum_{n=1}^{\infty} a_n n x^{n-1}}_{k=1} = f(x) + \varepsilon \cdot f'(x)$$

$\Rightarrow f(x+\varepsilon) - f(x) = \varepsilon \cdot f'(x)$

## Matrix Representation

$$a + b\varepsilon \mapsto \begin{pmatrix} a & b \\ 0 & a \end{pmatrix}$$

because  $\varepsilon = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow \varepsilon^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

## Application

observe:  $f \begin{pmatrix} x & 1 \\ 0 & x \end{pmatrix} = \begin{pmatrix} f(x) & f'(x) \\ 0 & f(x) \end{pmatrix}$

$$\downarrow \qquad \uparrow$$
$$f(x+\varepsilon) = f(x) + \varepsilon \cdot f'(x)$$

Recall

$$a + ib \mapsto \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

'cause  $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^2 = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$